

MOVIE TICKET BOOKING WEBSITE

Project Overview

The Movie Ticket Booking Website is a web-based platform designed to provide an online solution for discovering movies, booking movie tickets, and managing cinema screening information. The system provides a centralized platform through which customers can browse movies, view available cinemas and showtimes, select seats, complete online payment, and receive digital movie tickets.

The system provides detailed information about active and upcoming movies, including movie titles, genres, descriptions, durations, release dates, age ratings, posters, and trailers. Customers can search and filter movies, view movie details, select a location and cinema, choose an available showtime, view the corresponding screening-room seat layout, select seats, and complete the booking process.

The system manages information about customers, movies, locations, cinemas, screening rooms (halls), seat layouts, seats, showtimes, bookings, payment attempts, and tickets. A booking records the customer, selected movie and showtime, selected seats, price, booking status, and associated payment information.

The booking process includes a temporary seat-holding mechanism. After a customer confirms their seat selection, the selected seats are held for 10 minutes while the customer completes payment. Held seats cannot be selected by other customers. If payment is successfully confirmed, the seats become permanently booked and a digital ticket is generated. If the payment is not successfully completed within the required period, the held seats are released and become available again.

The system also provides an administration interface for managing movies, locations, cinemas, screening rooms, seat layouts, and showtimes. Administrators can create and edit movie information, configure venue information, assign predefined seat layouts to screening rooms, and create or remove valid showtimes according to the system's scheduling rules.

The system includes Cinema Staff functionality for validating customer tickets through QR codes at the cinema entrance.

The project is designed as a web application, with an emphasis on reliable and testable user workflows. Automated end-to-end testing using TestCafe will be used to test important customer, administrator, payment, seat-booking, and ticket-validation scenarios.

Main Functions

Customer Management

The system provides customer account management functionality.

Customers can:

- Register an account using their email address.
- Verify their email address during registration.
- Log in and log out of the system.
- Reset their password through their current verified email address.
- Update their personal profile information.
- Change their email address by verifying the new email address.
- Access only their own account, bookings, and tickets.

Only registered and authenticated customers can purchase movie tickets.

Movie Management

The system stores and displays movie information, including:

- Movie title
- Genre
- Duration
- Description
- Release date
- Age rating
- Poster
- Trailer
- Other configured movie information

Administrators can create and edit movie information.

A movie can be archived when it has no future showtimes. Archived movies are removed from the active movie catalog and cannot receive new showtimes, but their information is retained so that historical tickets and bookings remain valid.

Cinema and Screening Room Management

The system organizes cinemas using the following hierarchy:

Location → Cinema → Hall → Seat Layout → Seat

A Location represents a geographical area in Vietnam, such as a district or ward. Each Location can contain multiple Cinemas, and each Cinema can contain multiple Halls.

Each Hall is associated with a predefined seat layout. The seat layout represents the physical arrangement of seats in that Hall.

Once a Hall has been used by a Showtime, its seat layout cannot be changed in a way that would alter existing or historical seat information.

Showtime Management

The system manages movie screening schedules containing:

- Movie
- Location
- Cinema
- Hall
- Date
- Starting time

Administrators can create and manage showtimes.

The system validates showtime scheduling rules. A Hall cannot contain overlapping showtimes, while different Halls within the same Cinema may have simultaneous showtimes.

A required **15-minute buffer** is maintained between consecutive showtimes in the same Hall in addition to the movie's duration.

For example, if a movie begins at 19:00 and has a duration of 120 minutes, the next showtime in the same Hall cannot begin before 21:15.

The system rejects invalid showtimes and provides the administrator with an explanation of the scheduling conflict.

A showtime with no confirmed bookings can be removed. A showtime with one or more confirmed bookings cannot be removed.

Movie Search and Browsing

Customers can browse active and upcoming movies through the movie catalog.

The system provides search and filtering functions that allow customers to find movies based on attributes such as:

- Title
- Genre

- Release date
- Other supported movie information

Customers can select a movie to view its detailed information before choosing a showtime.

Archived movies are not shown as active movies in the normal catalog.

Showtime and Seat Selection

After selecting a movie, customers can choose a Location, Cinema, and Showtime.

The system then displays the corresponding Hall and its predefined seat layout.

Seats are represented according to their current state:

- **Available** — can be selected.
- **Held** — temporarily unavailable because another customer is completing payment.
- **Booked** — permanently unavailable because the ticket has been successfully purchased.

A showtime is considered **Sold Out only when all seats are booked**. Sold-out showtimes remain visible but are visually disabled and cannot be selected.

Ticket Booking

Customers can select available seats and confirm their selection.

When the confirmation request is processed, the system verifies the current availability of the requested seats. The system must not rely only on the seat state displayed in the customer's browser.

If the seats are available, the system places them in a temporary **10-minute hold**.

During the hold period:

- Other customers cannot select the held seats.
- The customer can proceed with payment.
- A failed payment does not immediately release the seats.
- When the hold expires without successful booking confirmation, the seats return to the Available state.
- If the booking session is explicitly terminated, the system can release the held seats.

The system also prevents concurrent customers from successfully reserving the same seat. When two customers attempt to confirm the same available seat at approximately the same time, the

first valid request received by the system obtains the seat, while subsequent conflicting requests are rejected.

Payment

The system integrates with an external **Payment Provider** to process online payments.

The application manages the booking and payment workflow but does not directly function as a bank or card-processing network.

Each booking may have multiple payment attempts, but **only one payment attempt can be active at a time**.

Customers can retry payment after a failed attempt while their seat hold is still active.

The system handles the following payment states:

- **Success** — the booking is confirmed and the selected seats become booked.
- **Failed** — the seats remain held until the original hold expires.
- **Cancelled** — the payment attempt stops and the seats are released.
- **Pending** — the seats remain held while the system determines the payment result.
- **Unknown** — the system queries the Payment Provider before deciding whether the payment succeeded or failed.

If a successful payment result arrives after the original seat hold has expired, the system checks whether all requested seats are still available. If they are unavailable because another customer has booked them, the system does not issue a ticket and requests the appropriate payment cancellation or refund through the Payment Provider.

The system must also prevent duplicate payment charges and duplicate ticket generation when the customer submits repeated payment requests.

Ticket Management

After successful payment confirmation, the system generates a digital movie ticket.

The ticket contains:

- Movie information
- Cinema
- Showtime
- Selected seats

- Customer name
- Booking ID
- QR code
- Ticket price

Customers cannot cancel a successfully purchased ticket under the current system rules.

Customers can access their upcoming tickets through the **My Tickets** section even when the tickets were purchased more than 30 days earlier.

Past booking history is displayed for the previous **30 days**. The 30-day rule controls normal history visibility; it does not mean that older booking records are deleted from the system.

Historical tickets remain associated with their original movie, showtime, cinema, hall, and seat information even after a movie has been archived.

Ticket Validation

Cinema Staff can use the ticket-validation function to verify customer tickets at the cinema entrance.

The system validates the ticket's QR code and checks whether:

- The ticket exists.
- The ticket is valid.
- The ticket belongs to the relevant showtime.
- The ticket has not already been used.

A successfully validated ticket is marked as **Used** so that it cannot be reused.

Administration

The administration interface provides centralized management of the system's movie and cinema-screening information.

Administrators can:

- Create and edit movie information.
- Manage the configured Locations, Cinemas, and Halls.
- Assign predefined seat layouts to Halls.
- Create and manage showtimes.

- Remove showtimes that do not have confirmed bookings.
- Archive movies when no future showtimes remain.

Administrator accounts are provisioned by the system and use role-based authorization. Customers cannot access administrator functions.

Security and Authorization

The system applies authentication and role-based authorization to protected functions.

The main roles are:

- **Customer**
- **Admin**
- **Cinema Staff**

Customers cannot access administrative functions, while Cinema Staff cannot perform administrative management operations.

Customer passwords must be stored securely and never as plaintext.

Payment-provider callbacks and payment results must be verified before the system accepts them. The system validates relevant information such as the payment reference, booking reference, amount, currency, transaction state, and provider authentication/signature.

System Interface

The system is implemented as a responsive **web application** using standard web technologies.

The frontend uses **HTML, CSS, and JavaScript**, with **React.js** used as the frontend application architecture.

The frontend communicates with the backend through web APIs. The backend provides application logic, authentication, authorization, booking management, seat-state management, payment integration, and ticket processing.

A database stores persistent system information, including customers, movies, venues, halls, seats, showtimes, bookings, payment attempts, and tickets.

The application consists of customer-facing, administrative, and ticket-validation interfaces that are accessed through a web browser.

Expected Outcome

The completed system provides a centralized web-based platform for online movie ticket booking and cinema screening management.

Customers can complete the main booking process through the website:

Register and verify account → browse movies → view movie details → choose location and cinema → choose showtime → select seats → hold seats → complete payment → receive digital ticket.

Administrators can manage movies, venues, seat layouts, and showtimes while the system enforces scheduling and booking constraints.

Cinema Staff can validate purchased tickets using QR codes at the cinema entrance.

The system also provides controlled handling of seat concurrency, temporary seat reservations, payment failures and uncertain payment states, authentication, authorization, and historical ticket information.

The resulting web application provides both a functional movie-ticket booking platform and a suitable environment for automated **end-to-end software testing using TestCafe**.